

Research framework: from an established monetization–success relationship to its contextual scope

Interaction analysis applied to assess **effect-measure modification**: whether measured Ads- and IAP–marketplace-success associations vary across game, market, and temporal contexts. Estimated on two scales — additive (DDP) and multiplicative (RoR) — each with 95% confidence intervals.

RESEARCH PROBLEM	WHAT THE FIELD HAS ESTABLISHED	→ GAP	WHY THAT IS NOT MERELY INCOMPLETE
	Ads and IAP relate to player experience and commercial outcomes, and games, players, and national markets differ from one another. This evidence is valid. It is reported as an overall relationship, or as differences <i>between</i> groups of games and players — and then read as guidance for games and markets it never described.		An aggregate estimate can hold while the relationship it reports is weaker, absent, or reversed in particular contexts within the same evidence. Evidence of this kind does not under-inform decisions — it misdirects them, with confidence . Group differences establish that games and players differ; they cannot establish that the focal relationship differs, so the misdirection is invisible to the analyses currently used to detect it. This study makes that variation the object of analysis.
			Groups differ observed heterogeneity

WHY INTERACTION IS NEEDED	Without examining variation within a relationship, valid overall evidence can still misdirect decisions. THE METHODOLOGICAL GAP IN GAME RESEARCH Research on game and marketplace success is predominantly predictive and explanatory: it identifies which features relate to or predict favourable outcomes. That is a property distinct from interaction structure — models can perform competitively on conventional prediction criteria without necessarily capturing interaction effects — so this body of evidence leaves unexamined whether the relationships supporting its performance remain equivalent across contexts.	ESTABLISHED IN EPIDEMIOLOGY, ADAPTED HERE Epidemiology treats this as a standard question: interaction and effect modification examine whether the effect of one exposure on an outcome differs across levels of another factor, since an intervention may benefit some subgroups while harming others, and an overall estimate can conceal both. The same concerns apply to game research, where these principles have not been systematically applied; this study adapts them to it.
	Feature → Outcome ≠ Feature → Outcome Context prediction / explanation conditional relationship	

THREE EVIDENTIAL LIMITATIONS — LONG-ESTABLISHED CONCERNS IN EPIDEMIOLOGY, NAMED HERE FOR GAME RESEARCH

False irrelevance Opposing relationships in different contexts offset one another, producing a weak or null overall estimate. $y \sim \tau = 0 \quad \text{while} \quad y \sim \tau \mid c_1 > 0 > y \sim \tau \mid c_2$ Consequence: a consequential practice is judged unimportant and dropped from consideration.	False stability A supported overall relationship is assumed to hold evenly across the contexts represented within the evidence. $y \sim \tau \mid c_1 = y \sim \tau \mid c_2$ assumed, not estimated Consequence: effort is directed at games and players for whom the relationship is weaker, absent, or reversed.	False generalisation Evidence is carried beyond the market, population, or period in which it was established. $y \sim \tau \mid C_{\text{observed}}$ applied to C_{target} without estimating it there Consequence: a practice is transferred to a market where the relationship no longer holds.
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THIS STUDY PROPOSES · SEVEN-STEP INTERACTION-ANALYSIS FRAMEWORK 95% CI on both scales	Contingency tables → Probabilities → Differences → Odds ratios → Additive · DDP → Multiplicative · RoR → Interpretation
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WHAT IS AT STAKE	In each case the original finding remains empirically valid; what is unknown is its contextual scope. Monetization decisions commit design, development, and marketing resources before their effects are observable — and the damage is often not simply expensive to undo. Sometimes irreversible Monetization is built into progression, economy, and content pacing; removing it can require rebuilding the game rather than reverting a setting.	Reputational damage persists Ratings, reviews, and community judgement outlast the change that caused them, and continue to shape acquisition after the design is corrected.	Distribution can be withdrawn Store policy enforcement or delisting can remove a title from the marketplace before a correction ships — foreclosing the opportunity to fix it at all.
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OBJECTIVE & QUESTION	RQ To develop a systematic and reproducible framework for establishing the contextual scope of empirical relationships in game research, making variation in the conditions under which a relationship applies an explicit component of the evidence itself.	RQ How do Ads- and IAP–marketplace-success relationships vary across game genres, and how consistently do these relationships recur across national markets and time? Addressed through five sub-questions, one per stage (RQ1–RQ.5 below).
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DATA & DESIGN	POPULATION Free-to-play Android games, Google Play — 1,778,700 game–market–month observations (GameMetaDroid)	FOCAL FEATURE τ Ads availability; IAP availability — analysed separately	CONTEXT c 10 genres (one-vs-rest); 42 national markets; 14 months, Jan 2025–Feb 2026	OUTCOME y Install success and rating success — at or above the country–month median	ESTIMATION 2×2 tables per stratum → conditional probabilities, odds ratios → DDP, RoR (95% CI) All estimates are associations within observational marketplace data; no causal effects are claimed.
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FIVE STAGES	1 Genre conditioning	2 Cross-market recurrence	3 Temporal stability	4 Forms of play	5 National conditions
	RQ.1 Does the overall relationship conceal variation across forms of play?	RQ.2 Do the conditional relationships recur within individual national markets?	RQ.3 Do recurring relationships persist as the marketplace evolves?	RQ.4 What play do the recurring positive and negative profiles represent?	RQ.5 What national characteristics are associated with cross-market variation in interaction magnitude?
	Game-studies research treats genre as more than a classificatory label, reflecting differences in the conventions, expectations, and structures through which gameplay is organised and experienced. Because Ads and IAP are implemented within different organisations of play, genre is a theoretically meaningful context for asking whether the marketplace relevance of the same mechanism varies across games. Conditioning on genre estimates the relationship within each, rather than comparing genres to one another.	Games ship the same design to different populations, so the national market is where one practice meets different conditions of access, payment, and reception. Despite substantial evidence of national variation in gaming and monetization, this form of cross-country validation of a monetization–success relationship has not been systematically examined.	Titles age, games update, catalogues turn over. Without a temporal test, recurrence could be a snapshot of one market state.	Genre establishes that forms of play matter; reading the games themselves specifies which properties of play the recurring profiles are composed of.	Markets differ in demographic, economic, developmental, and digital conditions, which may explain cross-market variation in monetization.
	Method estimation step $y \sim \tau \mid c$ 20 monetization × genre profiles, pooled; seven-step framework applied — DDP and RoR with 95% CI.	$y \sim \tau \mid c, C$ Framework re-applied within each of 42 markets; direction classified + / 0 / − / M (mixed or scale-specific).	$y \sim \tau \mid c, C, T$ Framework re-applied within country–month strata across 14 monthly periods.	profiles → forms of play 176 games from 9 recurring profiles, read from store and gameplay sources, coded for 8 characteristics of play.	DDP _c ~ Age, PPP, HDI, NET, DP Spearman p per profile–indicator pair (n = 40–42), Benjamini–Hochberg FDR, q < .05.
	Direction reverses by genre ADS + Puzzle .349 · Casual .262 ADS − RPG −.441 · Strategy −.363 IAP + RPG .397 · Strategy .340 · Simulation .185 IAP − Puzzle −.133 · Casual −.085 A qualitative interaction, not a weaker or stronger one: the overall estimate describes neither side.	Five profiles recur in every market Ads × Puzzle, Casual — 42 / 42 positive IAP × RPG, Strategy, Simulation — 42 / 42 positive IAP × Sports — 23 + · 6 null · 10 − · 3 mixed Recurrence is established for five profiles; one pooled positive estimate dissolves into opposing market directions.	And persist across 14 months Recurring profiles — 14 / 14 months consistent IAP × Sports — 10 + · 4 − across months The recurring relationships are not period-specific, so their reproducibility cannot be dismissed as a snapshot.	Opposed configurations of play ADS + Interruptible, bounded, self-contained play — natural pauses 49%, activity boundaries 79%, contained development 44% IAP + Persistent, accumulating, investment-carrying play — capability development 63%, accumulation 68%, state persistence 52% The statistical crossover corresponds to a crossover in how play is organised — the conditional relationship has a gameplay account, not only a label.	Cross-market variation is systematic IAP profiles strengthen with all five indicators — 15 / 15 significant; PPP $\rho = .751$ –.806 Ads × Puzzle is associated in the opposite direction — age $\rho = -.842$ The two mechanisms are ordered in opposite directions across national conditions. Recurrence is the weaker claim: the marketplace advantage associated with a monetization practice varies with where play circulates.

THE PARADOX	STAGES 2–3 ESTABLISH The relationship recurr s — 42 of 42 markets, 14 of 14 months, on both interaction scales.	<i>yet</i>	STAGES 4–5 ESTABLISH It does not carry equivalent significance — magnitude is ordered by forms of play and by market conditions.
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AND IT IS NOT AN OVERALL-LEVEL ARTEFACT — THE REVERSAL REAPPEARS AT EVERY LEVEL OF ANALYSIS

ACROSS GENRES One mechanism, opposite directions Ads: Puzzle + .349 · RPG −.441 . The pooled Ads estimate describes neither genre.	WITHIN ONE GENRE One form of play, opposite mechanisms RPG: Ads −.441 · IAP + .397 . The same games reward one monetization route and penalise the other.	WITHIN ONE PROFILE, ACROSS COUNTRIES One profile, opposite markets IAP × Sports is pooled-positive (+ .099) yet 23 positive · 6 null · 10 negative · 3 mixed across markets.	WITHIN RECURRING PROFILES One direction, unequal strength IAP × Strategy holds in all 42 markets while magnitude rises with PPP ($\rho = .800$); Ads × Puzzle falls with median age ($\rho = -.842$).
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Conditionality is therefore nested: it appears between genres, between mechanisms within a genre, between countries within a profile, and in magnitude within relationships that recur everywhere. An aggregate estimate can be reproduced at each level without describing any of them.

CONTRIBUTION	METHODOLOGICAL An Interaction-Analysis Framework for game research A reproducible seven-step pipeline adapting epidemiological principles of interaction and effect-measure modification, demonstrated here through the five stages below — so evidence about game success can state its own conditions of validity on additive and multiplicative scales.	EMPIRICAL The boundaries of monetization evidence Establishes the scope of Ads- and IAP–success relationships: where they hold, where they reverse, how far they travel across 42 markets and 14 months, and which structures of play and market conditions are associated with their strength — the evidence a monetization decision needs before it commits.	CONCEPTUAL The cultural paradox of success Separates reproducibility from transferability: a practice can succeed everywhere and be worth something different in each place. The paradox is cultural in the broad sense — constituted jointly by forms of play, national marketplaces, and historical moment, none of which an aggregate estimate represents.
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IMPLICATIONS
An aggregate finding — say, *this much ad exposure is fine for players* — becomes a contextual account: **in which forms of play** that holds, **in which markets** and for **how long** it recurs, and therefore a stronger basis for interpretation, generalisability, and decision-making.

Research	Game design & development	Publishing, market entry & organisations
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Scope becomes part of what a finding reports, and heterogeneity is reported as evidence rather than absorbed into an average. The same logic extends to retention, difficulty, and pricing, and to regression, machine-learning, and XAI settings.
AVOIDS Misleading generalisations, and weak external validity and limited replicability.

Monetization is matched to the structure of play that affords it — interruptible and bounded, or persistent and accumulating — and another title's evidence becomes usable on a stated condition: comparable organisation of play, not a comparable genre label.
AVOIDS Ineffective or harmful monetization strategies, committed where revision means rebuilt design rather than a reverted setting.

Markets comparable in demographic, socioeconomic, and digital conditions serve as reference points where target-market data is absent, and resources follow transfer certainty: act where relationships recur, validate where magnitude varies, experiment where direction is unstable.
AVOIDS Misallocation of resources, and poor strategic decisions with the opportunity costs they carry.